

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building **large-scale Node.js microservices** for high-growth products. Architected an event-driven **microservices backend** with Express/NestJS-style services on **Apache Kafka, Docker, and Kubernetes** for FIDA Uganda; led a zero-downtime migration of 20+ apps to Firebase/GCP at MTailor; and built real-time sync pipelines on Node.js + Pub/Sub. Deep **MongoDB**, RESTful-API, and containerized-deployment experience — well suited to owning backend systems for African mobility, on-demand, and fintech services.

CORE COMPETENCIES

Node.js

MongoDB

Express.js / NestJS

RESTful APIs

Microservices

Docker & Kubernetes

Scaling & Load Balancing

GCP / Firebase

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built **real-time two-way sync** between MongoDB and Firestore using **Node.js** and Google **Pub/Sub** for message processing, keeping a live system running throughout.
- Cut infrastructure cost by **\$5,000/month** by migrating all services off AWS; moved file storage from S3 to GCS via Python.
- Shipped a 3D visualisation feature (video overlay + ffmpeg) and Express Shipping (+\$40 revenue/order), improving buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built a migration pipeline reading from MongoDB and writing to Firestore after complex processing logic.
- Documented the new Firebase SDKs (Node.js & Python) for engineers and trained the Ops team on dashboards and backup procedures.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected a **microservices backend** for FIDA Uganda's case-management app using **Apache Kafka, Docker, and Kubernetes**.
- Built a USSD legal-aid service communicating with a **Node.js** backend via **gRPC**; led a team of 4 developers.
- Delivered case-management and paralegal-database systems plus a React Native mobile app with Firebase push notifications.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Optimised a PWA delivering **real-time** stock prices from a Firebase backend; queried **2M+ Firestore records** via scheduled jobs.
- Extended backend APIs with Firebase Cloud Functions and **Express.js** on Heroku.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using **Node.js** and Web3; delivered React UIs from Figma under tight timelines.

PROJECTS

Microservices Backend — FIDA Uganda

Architected an event-driven backend for a legal case-management platform serving NGOs at scale.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC

Marketplace Recommendation Engine

Collaborative + content-based filtering in MongoDB for service-provider matching — relevant to on-demand marketplaces.

Node.js · MongoDB

Real-time Pub/Sub Sync Pipeline

Live two-way MongoDB↔Firestore sync running under parallel production traffic during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Backend: Node.js, Express.js, NestJS, REST APIs, Microservices, gRPC **Data:** MongoDB, Firestore, SQL, Google Pub/Sub, Apache Kafka

Cloud / Infra: GCP, Firebase, AWS, Docker, Kubernetes, Linux **Languages:** JavaScript, TypeScript, Python, Go (intermediate)

Frontend: React, React Native, HTML, CSS **Spoken:** English, Kiswahili